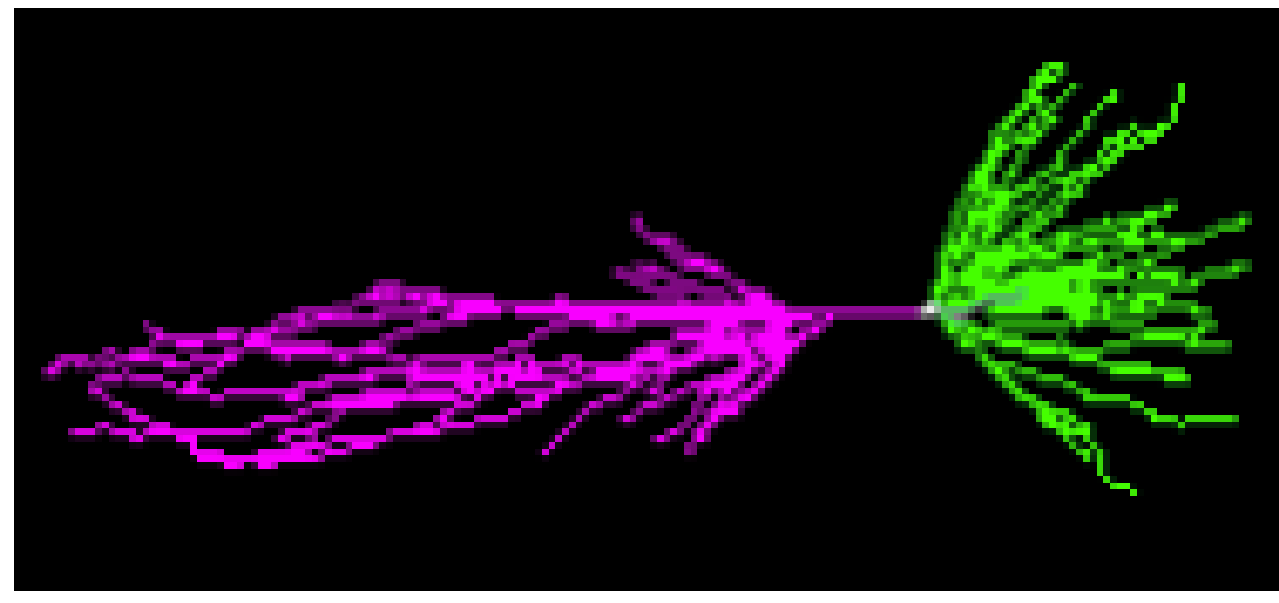


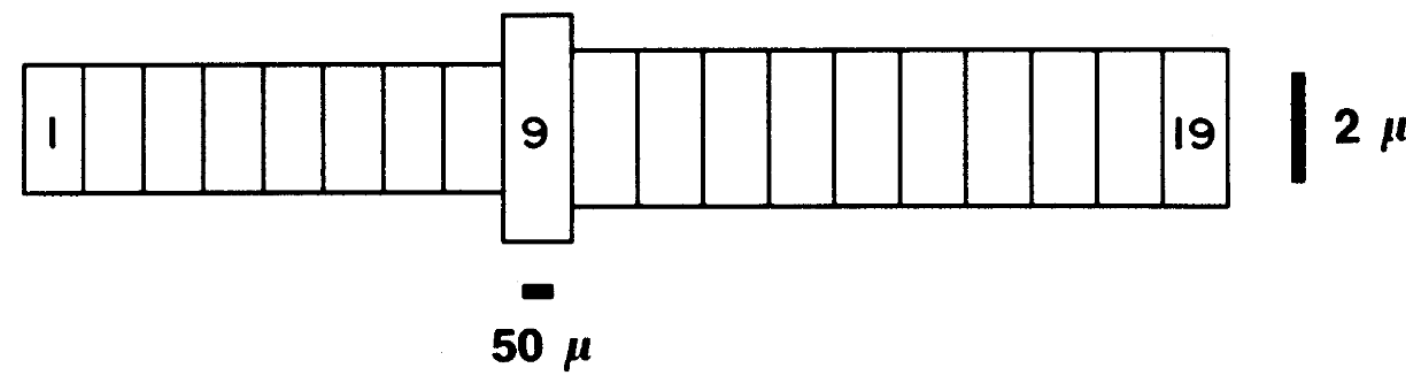
A Surrogate Model for Reducing the Computational Cost of Neuron Simulations

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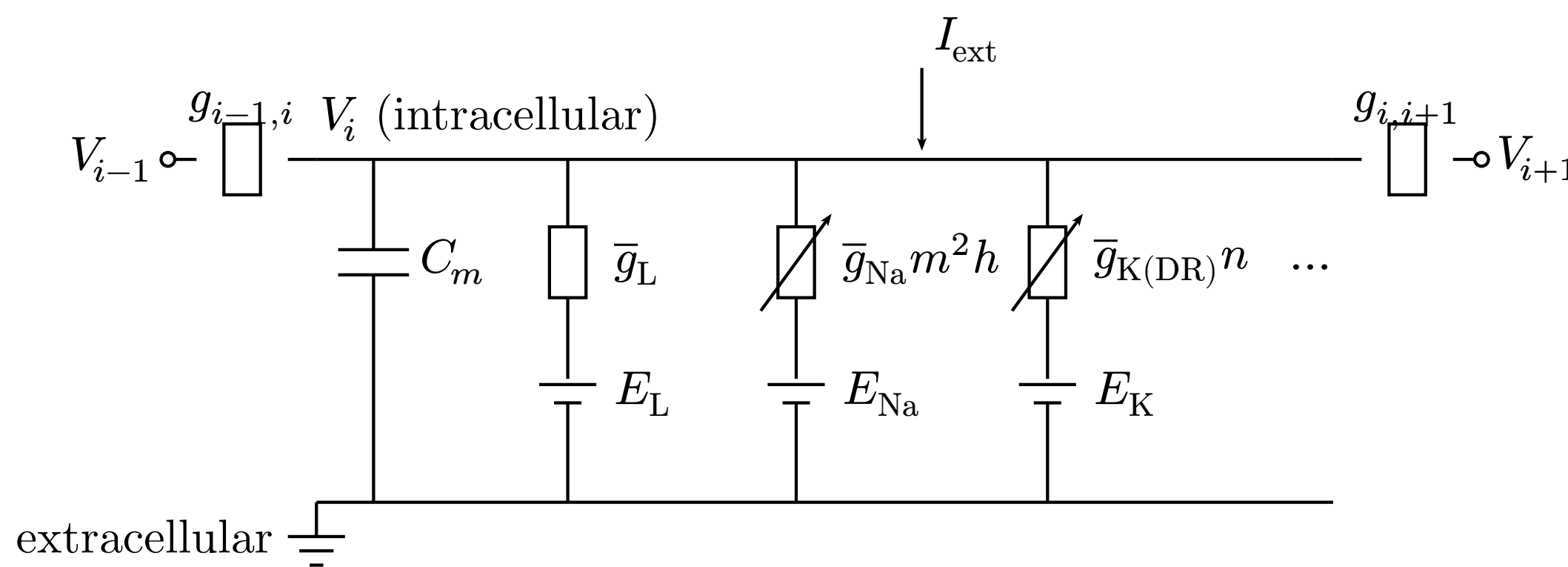
Introduction



guinea pig CA3 pyramidal neuron[1]



Modelled as **19 compartments** [2]



Equivalent circuit of compartment i
Each compartment: capacitance, **variable** ionic conductances, axial coupling $g_{i,i\pm 1}$.

Goal

Test whether **complex gate dynamics** remain tractable to a **simple regression** (SINDy) after a coordinate change — and thereby reproduce the membrane-potential response with **fewer state variables**.

Conductances depend on **gate variables**, ODEs

$$I_{Na} = \bar{g}_{Na} m^2 h (V - E_{Na})$$

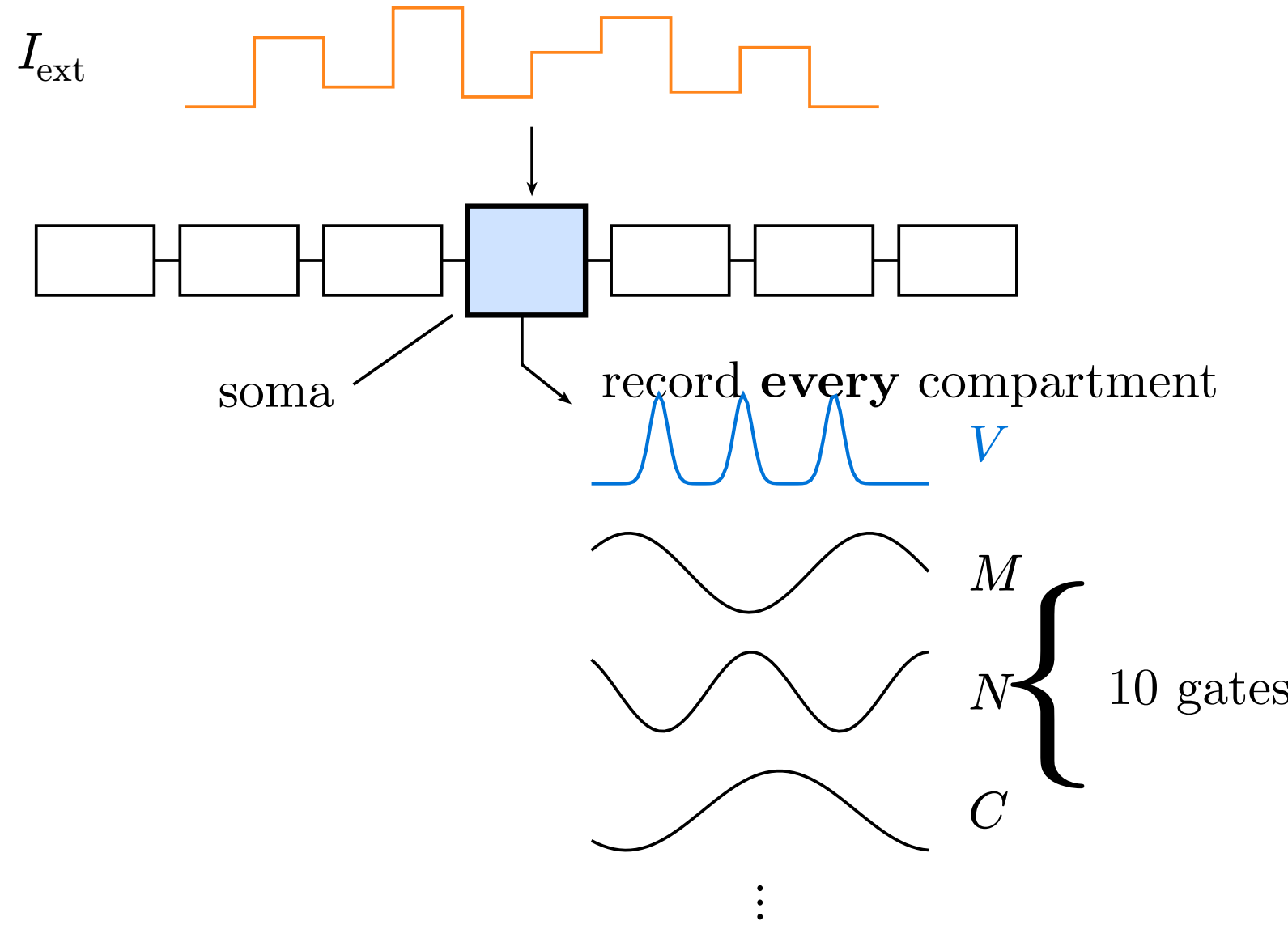
$$\frac{dm}{dt} = \alpha_m(V)(1 - m) - \beta_m(V)m$$

$$\alpha_m(V) = 0.32(13.1 - u) / (\exp((13.1 - u)/4) - 1)$$

→ **11 states per compartment** → **209** for 19; at brain scale the gates become a **memory bottleneck**.

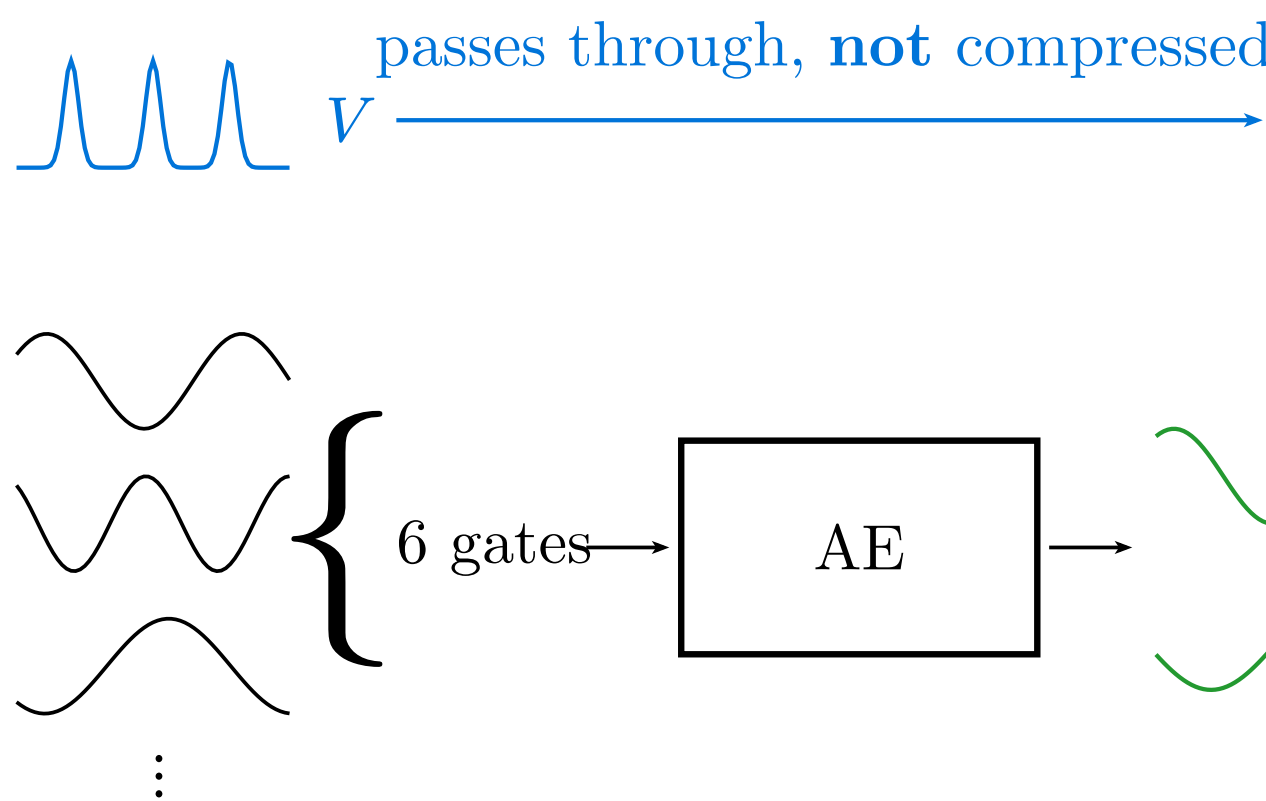
Methods

① Collect the training data



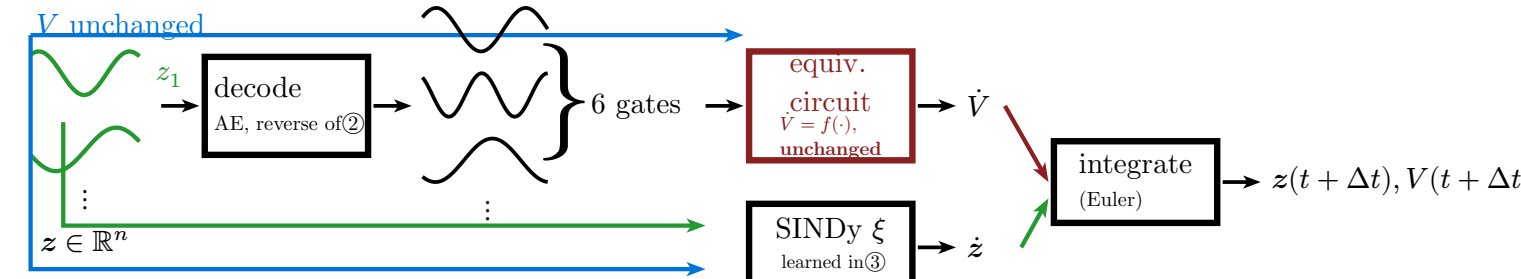
Inject a **random pulse train** at the soma; record V and **10 gates** for **all 19 compartments**.

② Compress gates



6 voltage-dependent gates ride a **low-dimensional manifold** → $z \in \mathbb{R}^n$, $n = 5$. V stays uncompressed.

④ Simulate: decoder-in-the-loop



③ Identify the latent ODEs

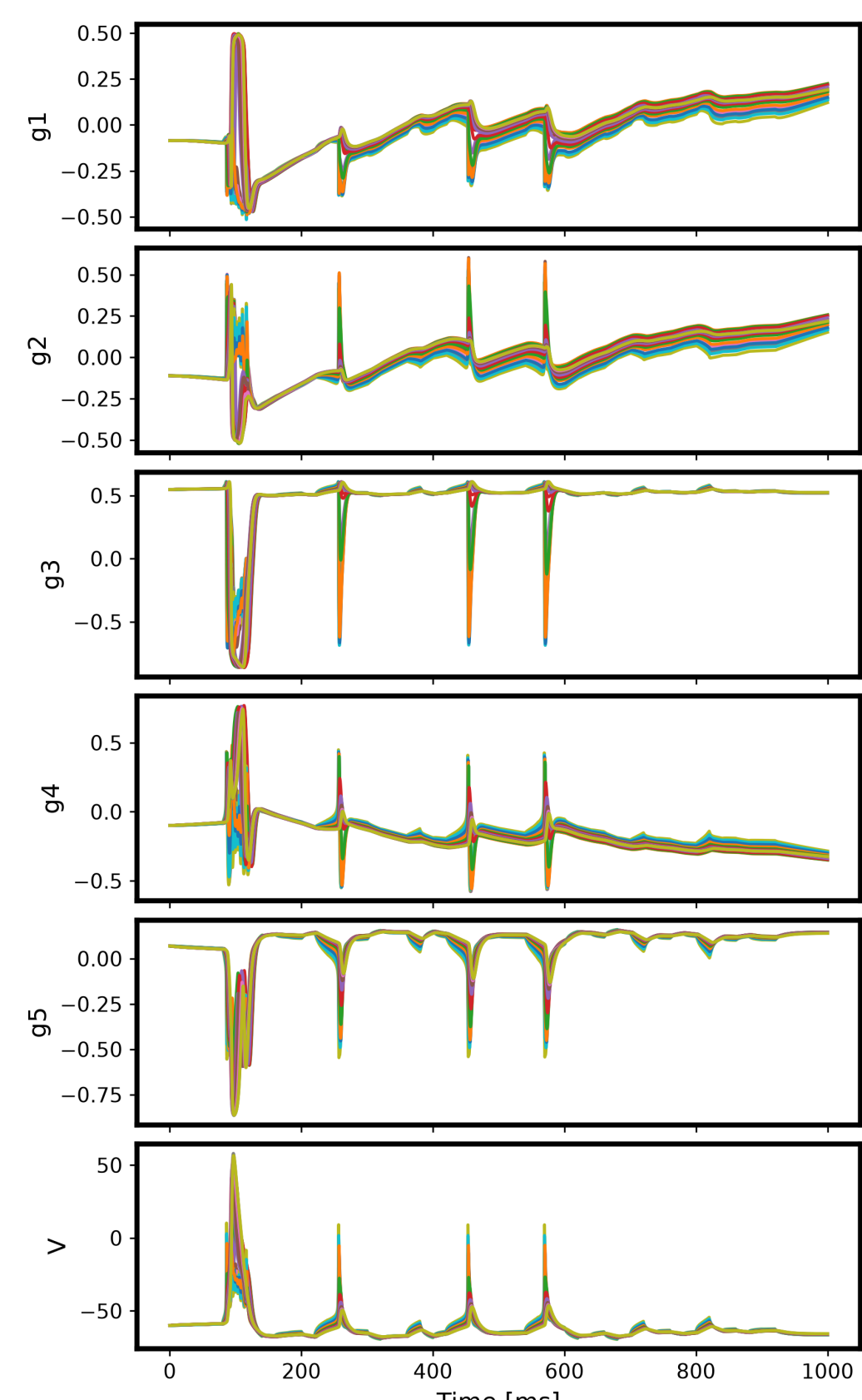
$$\begin{aligned} \frac{dz_1}{dt} &= \xi_{11} \alpha_M(V) + \xi_{12} \beta_M(V) z_1 + \dots \\ \frac{dz_2}{dt} &= \xi_{21} \alpha_N(V) + \dots \end{aligned}$$

SINDy [3] fits **only $\dot{z}$** ; $\dot{V}$ keeps **original physics**. Library is **physics-informed**: gates' own $\alpha(V), \beta(V)$, not a **41-term** generic one.

Each step: $z \rightarrow$ **decode** → gates feed the **unchanged** equivalent-circuit $\dot{V}$; SINDy updates z **in place** of the original gate ODEs.

Results and Discussion

Data



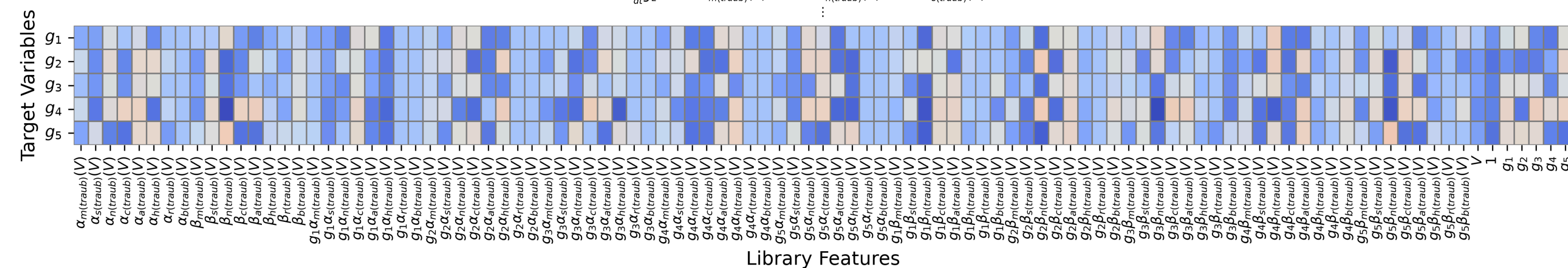
Latent trajectories used to fit the SINDy library.

④ Identified latent equations

SINDy Coefficients (SymLog Scale)

$$\hat{\beta}_1 = -2.116 \ln(\text{trunc}(V) + 56.54 \text{trunc}(V) - 7.64 \text{trunc}(V) + \dots$$

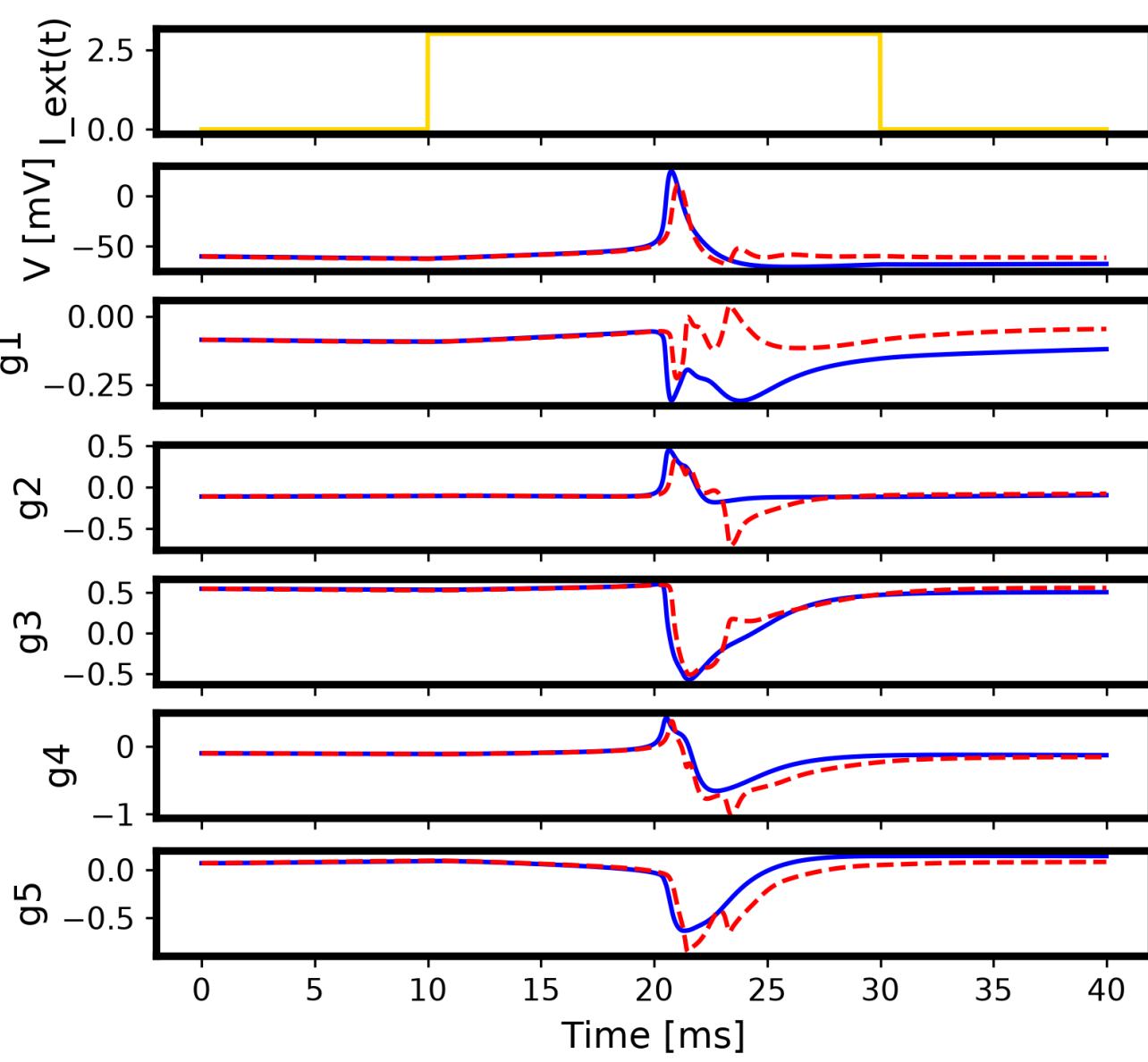
$$\hat{\beta}_2 = 6.928 \ln(\text{trunc}(V) + 238.02 \text{trunc}(V) - 51.84 \text{trunc}(V) + \dots$$



ξ over the physics-informed library.

- 79.6% non-zero** — accurate but **not sparse**.

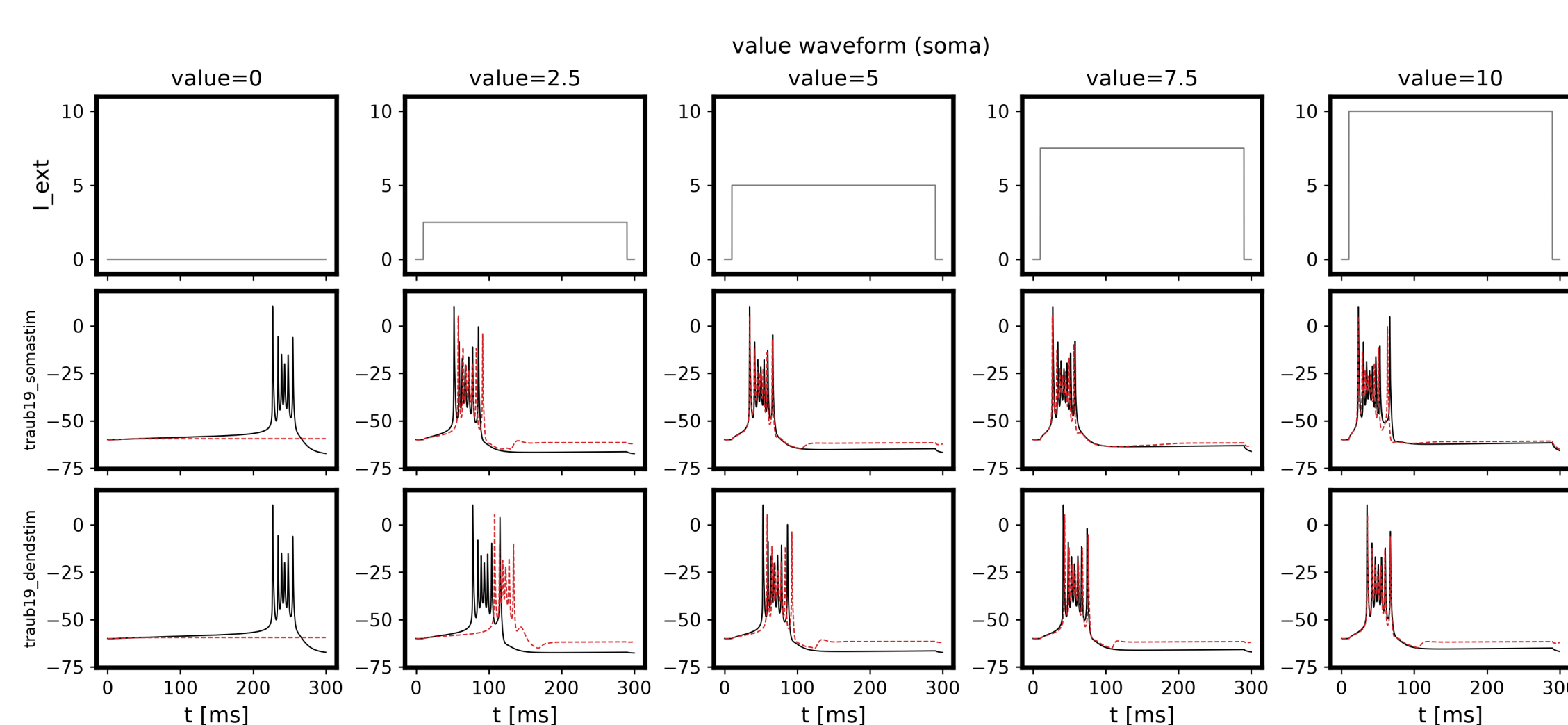
Action Potential comparison



20 ms, $3 \mu\text{A}/\text{cm}^2$ step: V and the **5 AE latents**.

- When it fires is accurate (latency err **0.3 ms**, AHP timing gap **3.1 ms**).
- How **big** is systematically **underestimated** — peak **13 mV** low (amplitude gap **12.9 mV**), rise / fall rate gap **21.0 / 10.3 mV/ms** — yet AHP depth gap is only **0.28 mV**.

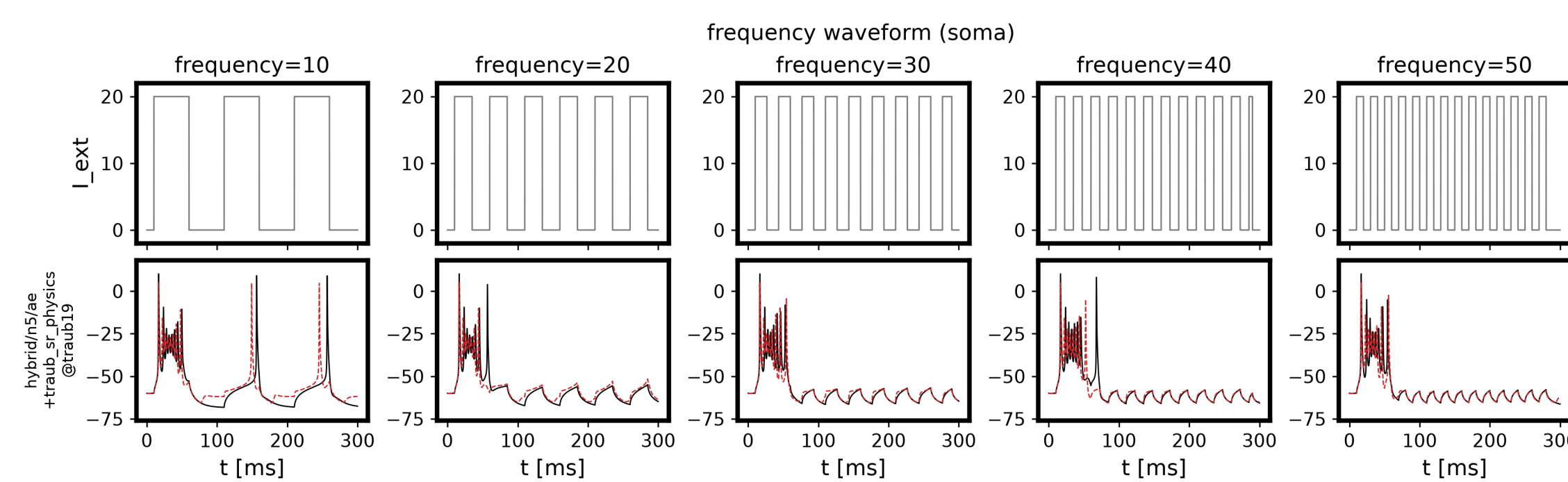
② New stimulus site



Amplitude sweep). **Top**: soma, as trained. **Bottom**: dendrite, **unseen**.

- Bursts reproduced at **both sites** for $I \geq 5$; fires **too early** near threshold.

③ Unseen periodic drive



Pulse train, 10–50 Hz — **outside training**.

- Matches for $f \geq 20$ Hz; **drops later spikes** at 10 Hz.

Conclusion

- Latent dynamics preserve **timing**.
- Error concentrates in the **fast, steep transition** — suggests the **AE encoding** ($n=5$), not the SINDy library, fails to preserve it.
- Replaced **all 19 compartments**, no divergence; transfers to **unseen site** • **frequency**.
- Core finding**: SINDy succeeds **not because we compressed** ($n: 6 \rightarrow 5$) — ⑤ shows the **same result uncompressed** ($n = 6$), so it is the **coordinate change itself** that makes gate dynamics learnable.
- Cost reduction follows **only** if compression matters — so it remains **unproven**: ξ is still dense (**79.6%** non-zero).
- **Future**: sparsify ξ , push n down, test on other channel models.

Code — github.com/MunechikaHaruki/SINDyNeuroSurrogate

[1] [Online]. Available: https://neuromorpho.org/neuron_info.jsp?neuron_name=L16a

[2] R. D. Traub, R. K. Wong, R. Miles, and H. Michelson, "A Model of a CA3 Hippocampal Pyramidal Neuron Incorporating Voltage-Clamp Data on Intrinsic Conductances," *Journal of Neurophysiology*, vol. 66, no. 2, pp. 635–650, Aug. 1991, doi: 10.1152/jn.1991.66.2.635.

[3] K. Champion, B. Lusch, J. N. Kutz, and S. L. Brunton, "Data-Driven Discovery of Coordinates and Governing Equations," *Proceedings of the National Academy of Sciences*, vol. 116, no. 45, pp. 22445–22451, Nov. 2019, doi: 10.1073/pnas.1906995116.

Conflicts of Interest — The authors declare no conflicts of interest regarding this manuscript.